

2520080012

OSSP PRACTICAL 9:

```
root@LAPTOP-0PA61QEN:~# ls/home
-bash: ls/home: No such file or directory
root@LAPTOP-0PA61QEN:~# su alurusaranya
alurusaranya@LAPTOP-0PA61QEN:~/root$ cd ~
alurusaranya@LAPTOP-0PA61QEN:~$ mkdir practical9
alurusaranya@LAPTOP-0PA61QEN:~$ cd practical9
alurusaranya@LAPTOP-0PA61QEN:~/practical9$ pwd
/home/alurusaranya/practical9
alurusaranya@LAPTOP-0PA61QEN:~/practical9$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            3.8G  0  3.8G  0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none            3.8G  4.0K  3.8G  1% /mnt/wsl
drivers          475G  149G  326G  32% /usr/lib/wsl/drivers
/dev/sdd        1007G  2.5G  954G  1% /
none            3.8G  32K  3.8G  1% /mnt/wslg
none            3.8G  0  3.8G  0% /usr/lib/wsl/lib
rootfs          3.8G  2.8M  3.8G  1% /init
none            3.8G  520K  3.8G  1% /run
none            3.8G  0  3.8G  0% /run/lock
none            3.8G  0  3.8G  0% /run/shm
none            3.8G  80K  3.8G  1% /mnt/wslg/versions.txt
none            3.8G  80K  3.8G  1% /mnt/wslg/doc
C:\             475G  149G  326G  32% /mnt/c
tmpfs           772M  20K  772M  1% /run/user/0
tmpfs           772M  20K  772M  1% /run/user/1000
alurusaranya@LAPTOP-0PA61QEN:~/practical9$ ls -la
total 8
drwxrwxr-x  2 alurusaranya alurusaranya 4096 Sep 28 04:47 .
drwxr-x--- 12 alurusaranya alurusaranya 4096 Sep 28 04:47 ..
alurusaranya@LAPTOP-0PA61QEN:~/practical9$ ls -li
total 0
alurusaranya@LAPTOP-0PA61QEN:~/practical9$ stat .
  File: .
  Size: 4096          Blocks: 8          IO Block: 4096   directory
Device: 8,48   Inode: 12083       Links: 2
Access: (0775/drwxrwxr-x)  Uid: ( 1000/alurusaranya)   Gid: ( 1000/alurusaranya)
Access: 2026-09-28 04:50:01.487770830 +0000

Access: 2026-09-28 04:50:01.487770830 +0000
Modify: 2026-09-28 04:47:07.309190877 +0000
Change: 2026-09-28 04:47:07.309190877 +0000
 Birth: 2026-09-28 04:47:07.309190877 +0000
alurusaranya@LAPTOP-0PA61QEN:~/practical9$ touch test.txt
alurusaranya@LAPTOP-0PA61QEN:~/practical9$ ls -li test.txt
12148 -rw-rw-r-- 1 alurusaranya alurusaranya 0 Sep 28 04:51 test.txt
alurusaranya@LAPTOP-0PA61QEN:~/practical9$ stat test.txt
  File: test.txt
  Size: 0             Blocks: 0          IO Block: 4096   regular empty file
Device: 8,48   Inode: 12148       Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/alurusaranya)   Gid: ( 1000/alurusaranya)
Access: 2026-09-28 04:51:47.599578191 +0000
Modify: 2026-09-28 04:51:47.599578191 +0000
Change: 2026-09-28 04:51:47.599578191 +0000
 Birth: 2026-09-28 04:51:47.599578191 +0000
alurusaranya@LAPTOP-0PA61QEN:~/practical9$ echo "Operating Systems Practical 9" > test.txt
alurusaranya@LAPTOP-0PA61QEN:~/practical9$ ls -li test.txt
12148 -rw-rw-r-- 1 alurusaranya alurusaranya 30 Sep 28 04:52 test.txt
alurusaranya@LAPTOP-0PA61QEN:~/practical9$ stat test.txt
  File: test.txt
  Size: 30            Blocks: 8          IO Block: 4096   regular file
Device: 8,48   Inode: 12148       Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/alurusaranya)   Gid: ( 1000/alurusaranya)
Access: 2026-09-28 04:51:47.599578191 +0000
Modify: 2026-09-28 04:52:07.599329799 +0000
Change: 2026-09-28 04:52:07.599329799 +0000
 Birth: 2026-09-28 04:51:47.599578191 +0000
alurusaranya@LAPTOP-0PA61QEN:~/practical9$ cat test.txt
Operating Systems Practical 9
alurusaranya@LAPTOP-0PA61QEN:~/practical9$ ls -la
total 12
drwxrwxr-x  2 alurusaranya alurusaranya 4096 Sep 28 04:51 .
drwxr-x--- 12 alurusaranya alurusaranya 4096 Sep 28 04:47 ..
-rw-rw-r--  1 alurusaranya alurusaranya  30 Sep 28 04:52 test.txt
alurusaranya@LAPTOP-0PA61QEN:~/practical9$ find . -maxdepth 2 -type f
./test.txt
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./test.txt
alurusaranya@LAPTOP-0PA61QEN:~/practical9$ df -T .
Filesystem      Type      1K-blocks    Used Available Use% Mounted on
/dev/sdd        ext4 1055762868 2544384 999515012   1% /
alurusaranya@LAPTOP-0PA61QEN:~/practical9$ findmnt .
alurusaranya@LAPTOP-0PA61QEN:~/practical9$ mount | head -20
none on /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2 type overlay (rw,nosuid,nodev,noatime,lowerdir=/modules,upperdir=/lib/modules/6.18.33.2-microsoft-standard-WSL2/rw/upper,workdir=/lib/modules/6.18.33.2-microsoft-standard-WSL2/rw/work,uuid=on)
none on /mnt/wsl type tmpfs (rw,relatime)
drivers on /usr/lib/wsl/drivers type 9p (ro,nosuid,nodev,noatime,aname=drivers;fmask=222;dmask=222,cache=0x5,access=client,msize=65536,trans=fd,rfd=8,wfd=8)
/dev/sdd on / type ext4 (rw,relatime,discard,errors=remount-ro,data=ordered)
none on /mnt/wslg type tmpfs (rw,relatime)
/dev/sdd on /mnt/wslg/distro type ext4 (ro,relatime,discard,errors=remount-ro,data=ordered)
none on /usr/lib/wsl/lib type overlay (rw,nosuid,nodev,noatime,lowerdir=/gpu_lib_packaged:/gpu_lib_inbox,upperdir=/gpu_lib/rw/upper,workdir=/gpu_lib/rw/work,uuid=on)
rootfs on /init type rootfs (ro,size=3942848k,nr_inodes=985712)
none on /dev type devtmpfs (rw,nosuid,relatime,size=3942848k,nr_inodes=985712,mode=755)
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,noatime)
proc on /proc type proc (rw,nosuid,nodev,noexec,noatime)
devpts on /dev/pts type devpts (rw,nosuid,noexec,noatime,gid=5,mode=620,ptmxmode=000)
none on /run type tmpfs (rw,nosuid,nodev,mode=755)
none on /run/lock type tmpfs (rw,nosuid,nodev,noexec,noatime)
none on /run/shm type tmpfs (rw,nosuid,nodev,noatime)
none on /dev/shm type tmpfs (rw,nosuid,nodev,noatime)
none on /run/user type tmpfs (rw,nosuid,nodev,noexec,noatime,mode=755)
binfmt_misc on /proc/sys/fs/binfmt_misc type binfmt_misc (rw,relatime)
cgroup2 on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime,nsdelegate)
none on /mnt/wslg/versions.txt type overlay (rw,relatime,lowerdir=/systemvhd,upperdir=/system/rw/upper,workdir=/system/rw/work,uuid=on)
alurusaranya@LAPTOP-0PA61QEN:~/practical9$ cat /proc/mounts | head -20
none /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2 overlay rw,nosuid,nodev,noatime,lowerdir=/modules,upperdir=/lib/modules/6.18.33.2-microsoft-standard-WSL2/rw/upper,workdir=/lib/modules/6.18.33.2-microsoft-standard-WSL2/rw/work,uuid=on 0 0
none /mnt/wsl tmpfs rw,relatime 0 0
drivers /usr/lib/wsl/drivers 9p ro,nosuid,nodev,noatime,aname=drivers;fmask=222;dmask=222,cache=0x5,access=client,msize=65536,trans=fd,rfd=8,wfd=8 0 0
/dev/sdd / ext4 rw,relatime,discard,errors=remount-ro,data=ordered 0 0

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d,rfd=8,wfd=8 0 0
/dev/sdd / ext4 rw,relatime,discard,errors=remount-ro,data=ordered 0 0
none /mnt/wslg tmpfs rw,relatime 0 0
/dev/sdd /mnt/wslg/distro ext4 ro,relatime,discard,errors=remount-ro,data=ordered 0 0
none /usr/lib/wsl/lib overlay rw,nosuid,nodev,noatime,lowerdir=/gpu_lib_packaged:/gpu_lib_inbox,upperdir=/gpu_lib/rw/upper,workdir=/gpu_lib/rw/work,uuid=on 0 0
rootfs /init rootfs ro,size=3942848k,nr_inodes=985712 0 0
none /dev devtmpfs rw,nosuid,relatime,size=3942848k,nr_inodes=985712,mode=755 0 0
sysfs /sys sysfs rw,nosuid,nodev,noexec,noatime 0 0
proc /proc proc rw,nosuid,nodev,noexec,noatime 0 0
devpts /dev/pts devpts rw,nosuid,noexec,noatime,gid=5,mode=620,ptmxmode=000 0 0
none /run tmpfs rw,nosuid,nodev,mode=755 0 0
none /run/lock tmpfs rw,nosuid,nodev,noexec,noatime 0 0
none /run/shm tmpfs rw,nosuid,nodev,noatime 0 0
none /dev/shm tmpfs rw,nosuid,nodev,noatime 0 0
none /run/user tmpfs rw,nosuid,nodev,noexec,noatime,mode=755 0 0
binfmt_misc /proc/sys/fs/binfmt_misc binfmt_misc rw,relatime 0 0
cgroup2 /sys/fs/cgroup cgroup2 rw,nosuid,nodev,noexec,relatime,nsdelegate 0 0
none /mnt/wslg/versions.txt overlay rw,relatime,lowerdir=/systemvhd,upperdir=/system/rw/upper,workdir=/system/rw/work,uuid=on 0 0
alurusaranya@LAPTOP-0PA61QEN:~/practical9$ df -hT
Filesystem      Type      Size  Used Avail Use% Mounted on
none            overlay   3.8G  0  3.8G  0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none            tmpfs     3.8G  4.0K  3.8G  1% /mnt/wsl
drivers          9p        475G  149G  326G  32% /usr/lib/wsl/drivers
/dev/sdd        ext4      1007G  2.5G  954G  1% /
none            tmpfs     3.8G  36K  3.8G  1% /mnt/wslg
none            overlay   3.8G  0  3.8G  0% /usr/lib/wsl/lib
rootfs          rootfs    3.8G  2.8M  3.8G  1% /init
none            tmpfs     3.8G  520K  3.8G  1% /run
none            tmpfs     3.8G  0  3.8G  0% /run/lock
none            tmpfs     3.8G  0  3.8G  0% /run/shm
none            overlay   3.8G  80K  3.8G  1% /mnt/wslg/versions.txt
none            overlay   3.8G  80K  3.8G  1% /mnt/wslg/doc
C:\             9p        475G  149G  326G  32% /mnt/c
tmpfs           tmpfs     772M  20K  772M  1% /run/user/0
tmpfs           tmpfs     772M  20K  772M  1% /run/user/1000
alurusaranya@LAPTOP-0PA61QEN:~/practical9$ cat /proc/filesystems

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alurusaranya@LAPTOP-0PA61QEN:~/practical9$ cat /proc/filesystems
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```
nodev sysfs
nodev tmpfs
nodev bdev
nodev proc
nodev cgroup
nodev cgroup2
nodev cpuset
nodev devtmpfs
nodev binfmt_misc
nodev configfs
nodev debugfs
nodev tracefs
nodev securityfs
nodev sockfs
nodev bpf
nodev pipefs
nodev ramfs
nodev hugetlbfs
nodev devpts
nodev ext3
nodev ext4
nodev ext2
nodev squashfs
nodev vfat
nodev fuseblk
nodev fuse
nodev fusectl
nodev virtiofs
nodev overlay
nodev udf
nodev xfs
nodev 9p
nodev efivarfs
nodev mqueue
nodev selinuxfs
nodev pstore
```

```
nodev pstore
nodev autofs
```

```
alurusaranya@LAPTOP-0PA61QEN:~/practical9$ cat /proc/mounts | grep -E "ext4|tmpfs"
```

```
none /mnt/wsl tmpfs rw,relatime 0 0
/dev/sdd / ext4 rw,relatime,discard,errors=remount-ro,data=ordered 0 0
none /mnt/wslg tmpfs rw,relatime 0 0
/dev/sdd /mnt/wslg/distro ext4 ro,relatime,discard,errors=remount-ro,data=ordered 0 0
none /dev/devtmpfs rw,nosuid,relatime,size=3942848k,nr_inodes=985712,mode=755 0 0
none /run tmpfs rw,nosuid,nodev,mode=755 0 0
none /run/lock tmpfs rw,nosuid,nodev,noexec,noatime 0 0
none /run/shm tmpfs rw,nosuid,nodev,noatime 0 0
none /dev/shm tmpfs rw,nosuid,nodev,noatime 0 0
none /run/user tmpfs rw,nosuid,nodev,noexec,noatime,mode=755 0 0
none /tmp/.X11-unix tmpfs ro,relatime 0 0
none /run/user tmpfs rw,relatime 0 0
tmpfs /run/user/0 tmpfs rw,nosuid,nodev,relatime,size=789720k,nr_inodes=197430,mode=700 0 0
tmpfs /mnt/wslg/run/user/0 tmpfs rw,nosuid,nodev,relatime,size=789720k,nr_inodes=197430,mode=700 0 0
tmpfs /run/user/1000 tmpfs rw,nosuid,nodev,relatime,size=789720k,nr_inodes=197430,mode=700,uid=1000,gid=1000 0 0
tmpfs /mnt/wslg/run/user/1000 tmpfs rw,nosuid,nodev,relatime,size=789720k,nr_inodes=197430,mode=700,uid=1000,gid=1000 0 0
```

```
alurusaranya@LAPTOP-0PA61QEN:~/practical9$ ls -l /dev
```

```
total 0
crw-r--r-- 1 root root    10, 235 Sep 28 04:46 autofs
drwxr-xr-x 2 root root    600 Sep 28 04:46 block
drwxr-xr-x 2 root root    120 Sep 28 04:46 bsg
crw----- 1 root root    10, 234 Sep 28 04:46 btrfs-control
drwxr-xr-x 2 root root   2860 Sep 28 04:46 char
crw----- 1 root root     5,   1 Sep 28 04:46 console
lrwxrwxrwx 1 root root    11 Sep 28 04:46 core -> /proc/kcore
crw----- 1 root root    10, 260 Sep 28 04:46 cpu_dma_latency
crw----- 1 root root    10, 203 Sep 28 04:46 cuse
drwxr-xr-x 6 root root    120 Sep 28 04:46 disk
crw-rw-rw- 1 root root    10, 258 Sep 28 04:46 dxg
lrwxrwxrwx 1 root root    13 Sep 28 04:46 fd -> /proc/self/fd
crw-rw-rw- 1 root root     1,   7 Sep 28 04:46 full
crw-rw-rw- 1 root root    10, 229 Sep 28 04:46 fuse
crw----- 1 root root    10, 228 Sep 28 04:46 hpet
drwxr-xr-x 2 root root     0 Sep 28 04:46 hugepages
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crw----- 1 root root 229, 0 Sep 28 04:46 hvc0
crw----- 1 root root 229, 1 Sep 28 04:46 hvc1
crw--w---- 1 root tty 229, 2 Sep 28 04:46 hvc2
crw----- 1 root root 229, 3 Sep 28 04:46 hvc3
crw----- 1 root root 229, 4 Sep 28 04:46 hvc4
crw----- 1 root root 229, 5 Sep 28 04:46 hvc5
crw----- 1 root root 229, 6 Sep 28 04:46 hvc6
crw----- 1 root root 229, 7 Sep 28 04:46 hvc7
crw----- 1 root root 10, 183 Sep 28 04:46 hwrng
lrwxrwxrwx 1 root root 12 Sep 28 04:46 initctl -> /run/initctl
crw-r--r-- 1 root root 1, 11 Sep 28 04:46 kmsg
crw-rw---- 1 root kvm 10, 232 Sep 28 04:46 kvm
lrwxrwxrwx 1 root root 28 Sep 28 04:46 log -> /run/systemd/journal/dev-log
crw-rw---- 1 root disk 10, 237 Sep 28 04:46 loop-control
brw-rw---- 1 root disk 7, 0 Sep 28 04:46 loop0
brw-rw---- 1 root disk 7, 1 Sep 28 04:46 loop1
brw-rw---- 1 root disk 7, 2 Sep 28 04:46 loop2
brw-rw---- 1 root disk 7, 3 Sep 28 04:46 loop3
brw-rw---- 1 root disk 7, 4 Sep 28 04:46 loop4
brw-rw---- 1 root disk 7, 5 Sep 28 04:46 loop5
brw-rw---- 1 root disk 7, 6 Sep 28 04:46 loop6
brw-rw---- 1 root disk 7, 7 Sep 28 04:46 loop7
drwxr-xr-x 2 root root 60 Sep 28 04:46 mapper
crw----- 1 root root 10, 227 Sep 28 04:46 mcelog
crw-r----- 1 root kmem 1, 1 Sep 28 04:46 mem
crw----- 1 root root 10, 220 Sep 28 04:46 mptctl
drwxrwxrwt 2 root root 40 Sep 28 04:46 mqueue
drwxr-xr-x 2 root root 60 Sep 28 04:46 net
crw-rw-rw- 1 root root 1, 3 Sep 28 04:46 null
crw----- 1 root root 10, 144 Sep 28 04:46 nvram
crw----- 108, 0 Sep 28 04:46 ppp
crw-rw-rw- 1 root tty 5, 2 Sep 28 04:54 ptmx
crw----- 1 root root 246, 0 Sep 28 04:46 ptp0
lrwxrwxrwx 1 root root 4 Sep 28 04:46 ptp_hyperv -> ptp0
drwxr-xr-x 2 root root 0 Sep 28 04:46 pts
brw-rw---- 1 root disk 1, 0 Sep 28 04:46 ram0
brw-rw---- 1 root disk 1, 1 Sep 28 04:46 ram1

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crw----- 1 root root 246, 0 Sep 28 04:46 ptp0
lrwxrwxrwx 1 root root 4 Sep 28 04:46 ptp_hyperv -> ptp0
drwxr-xr-x 2 root root 0 Sep 28 04:46 pts
brw-rw---- 1 root disk 1, 0 Sep 28 04:46 ram0
brw-rw---- 1 root disk 1, 1 Sep 28 04:46 ram1
brw-rw---- 1 root disk 1, 10 Sep 28 04:46 ram10
brw-rw---- 1 root disk 1, 11 Sep 28 04:46 ram11
brw-rw---- 1 root disk 1, 12 Sep 28 04:46 ram12
brw-rw---- 1 root disk 1, 13 Sep 28 04:46 ram13
brw-rw---- 1 root disk 1, 14 Sep 28 04:46 ram14
brw-rw---- 1 root disk 1, 15 Sep 28 04:46 ram15
brw-rw---- 1 root disk 1, 2 Sep 28 04:46 ram2
brw-rw---- 1 root disk 1, 3 Sep 28 04:46 ram3
brw-rw---- 1 root disk 1, 4 Sep 28 04:46 ram4
brw-rw---- 1 root disk 1, 5 Sep 28 04:46 ram5
brw-rw---- 1 root disk 1, 6 Sep 28 04:46 ram6
brw-rw---- 1 root disk 1, 7 Sep 28 04:46 ram7
brw-rw---- 1 root disk 1, 8 Sep 28 04:46 ram8
brw-rw---- 1 root disk 1, 9 Sep 28 04:46 ram9
crw-rw-rw- 1 root root 1, 8 Sep 28 04:46 random
lrwxrwxrwx 1 root root 4 Sep 28 04:46 rtc -> rtc0
crw----- 1 root root 248, 0 Sep 28 04:46 rtc0
brw-rw---- 1 root disk 8, 0 Sep 28 04:46 sda
brw-rw---- 1 root disk 8, 16 Sep 28 04:46 sdb
brw-rw---- 1 root disk 8, 32 Sep 28 04:46 sdc
brw-rw---- 1 root disk 8, 48 Sep 28 04:46 sdd
crw-rw---- 1 root disk 21, 0 Sep 28 04:46 sg0
crw-rw---- 1 root disk 21, 1 Sep 28 04:46 sg1
crw-rw---- 1 root disk 21, 2 Sep 28 04:46 sg2
crw-rw---- 1 root disk 21, 3 Sep 28 04:46 sg3
drwxrwxrwt 2 root root 40 Sep 28 04:46 shm
crw----- 1 root root 10, 231 Sep 28 04:46 snapshot
drwxr-xr-x 2 root root 60 Sep 28 04:46 snd
lrwxrwxrwx 1 root root 15 Sep 28 04:46 stderr -> /proc/self/fd/2
lrwxrwxrwx 1 root root 15 Sep 28 04:46 stdin -> /proc/self/fd/0
lrwxrwxrwx 1 root root 15 Sep 28 04:46 stdout -> /proc/self/fd/1
crw-rw-rw- 1 root tty 5, 0 Sep 28 04:46 tty

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crw-rw-rw- 1 root tty      5,  0 Sep 28 04:46 tty
crw--w---- 1 root tty      4,  0 Sep 28 04:46 tty0
crw--w---- 1 root tty      4,  1 Sep 28 04:46 tty1
crw--w---- 1 root tty      4, 10 Sep 28 04:46 tty10
crw--w---- 1 root tty      4, 11 Sep 28 04:46 tty11
crw--w---- 1 root tty      4, 12 Sep 28 04:46 tty12
crw--w---- 1 root tty      4, 13 Sep 28 04:46 tty13
crw--w---- 1 root tty      4, 14 Sep 28 04:46 tty14
crw--w---- 1 root tty      4, 15 Sep 28 04:46 tty15
crw--w---- 1 root tty      4, 16 Sep 28 04:46 tty16
crw--w---- 1 root tty      4, 17 Sep 28 04:46 tty17
crw--w---- 1 root tty      4, 18 Sep 28 04:46 tty18
crw--w---- 1 root tty      4, 19 Sep 28 04:46 tty19
crw--w---- 1 root tty      4,  2 Sep 28 04:46 tty2
crw--w---- 1 root tty      4, 20 Sep 28 04:46 tty20
crw--w---- 1 root tty      4, 21 Sep 28 04:46 tty21
crw--w---- 1 root tty      4, 22 Sep 28 04:46 tty22
crw--w---- 1 root tty      4, 23 Sep 28 04:46 tty23
crw--w---- 1 root tty      4, 24 Sep 28 04:46 tty24
crw--w---- 1 root tty      4, 25 Sep 28 04:46 tty25
crw--w---- 1 root tty      4, 26 Sep 28 04:46 tty26
crw--w---- 1 root tty      4, 27 Sep 28 04:46 tty27
crw--w---- 1 root tty      4, 28 Sep 28 04:46 tty28
crw--w---- 1 root tty      4, 29 Sep 28 04:46 tty29
crw--w---- 1 root tty      4,  3 Sep 28 04:46 tty3
crw--w---- 1 root tty      4, 30 Sep 28 04:46 tty30
crw--w---- 1 root tty      4, 31 Sep 28 04:46 tty31
crw--w---- 1 root tty      4, 32 Sep 28 04:46 tty32
crw--w---- 1 root tty      4, 33 Sep 28 04:46 tty33
crw--w---- 1 root tty      4, 34 Sep 28 04:46 tty34
crw--w---- 1 root tty      4, 35 Sep 28 04:46 tty35
crw--w---- 1 root tty      4, 36 Sep 28 04:46 tty36
crw--w---- 1 root tty      4, 37 Sep 28 04:46 tty37
crw--w---- 1 root tty      4, 38 Sep 28 04:46 tty38
crw--w---- 1 root tty      4, 39 Sep 28 04:46 tty39
crw--w---- 1 root tty      4,  4 Sep 28 04:46 tty4
crw--w---- 1 root tty      4, 40 Sep 28 04:46 tty40

crw--w---- 1 root tty      4, 48 Sep 28 04:46 tty48
crw--w---- 1 root tty      4, 49 Sep 28 04:46 tty49
crw--w---- 1 root tty      4,  5 Sep 28 04:46 tty5
crw--w---- 1 root tty      4, 50 Sep 28 04:46 tty50
crw--w---- 1 root tty      4, 51 Sep 28 04:46 tty51
crw--w---- 1 root tty      4, 52 Sep 28 04:46 tty52
crw--w---- 1 root tty      4, 53 Sep 28 04:46 tty53
crw--w---- 1 root tty      4, 54 Sep 28 04:46 tty54
crw--w---- 1 root tty      4, 55 Sep 28 04:46 tty55
crw--w---- 1 root tty      4, 56 Sep 28 04:46 tty56
crw--w---- 1 root tty      4, 57 Sep 28 04:46 tty57
crw--w---- 1 root tty      4, 58 Sep 28 04:46 tty58
crw--w---- 1 root tty      4, 59 Sep 28 04:46 tty59
crw--w---- 1 root tty      4,  6 Sep 28 04:46 tty6
crw--w---- 1 root tty      4, 60 Sep 28 04:46 tty60
crw--w---- 1 root tty      4, 61 Sep 28 04:46 tty61
crw--w---- 1 root tty      4, 62 Sep 28 04:46 tty62
crw--w---- 1 root tty      4, 63 Sep 28 04:46 tty63
crw--w---- 1 root tty      4,  7 Sep 28 04:46 tty7
crw--w---- 1 root tty      4,  8 Sep 28 04:46 tty8
crw--w---- 1 root tty      4,  9 Sep 28 04:46 tty9
crw-rw---- 1 root dialout  4, 64 Sep 28 04:46 ttyS0
crw-rw---- 1 root dialout  4, 65 Sep 28 04:46 ttyS1
crw-rw---- 1 root dialout  4, 66 Sep 28 04:46 ttyS2
crw-rw---- 1 root dialout  4, 67 Sep 28 04:46 ttyS3
crw-rw---- 1 root dialout  4, 68 Sep 28 04:46 ttyS4
crw-rw---- 1 root dialout  4, 69 Sep 28 04:46 ttyS5
crw-rw---- 1 root dialout  4, 70 Sep 28 04:46 ttyS6
crw-rw---- 1 root dialout  4, 71 Sep 28 04:46 ttyS7
crw----- 1 root root    10, 223 Sep 28 04:46 uinput
crw-rw-rw- 1 root root      1,  9 Sep 28 04:46 urandom
crw----- 1 root root    10, 257 Sep 28 04:46 userfaultfd
crw-rw---- 1 root tty      7,  0 Sep 28 04:46 vcs
crw-rw---- 1 root tty      7,  1 Sep 28 04:46 vcs1
crw-rw---- 1 root tty      7,  2 Sep 28 04:46 vcs2
crw-rw---- 1 root tty      7,  3 Sep 28 04:46 vcs3
crw-rw---- 1 root tty      7,  4 Sep 28 04:46 vcs4
```

```

crw-rw---- 1 root dialout 4, 70 Sep 28 04:46 ttyS6
crw-rw---- 1 root dialout 4, 71 Sep 28 04:46 ttyS7
crw-rw---- 1 root root 10, 223 Sep 28 04:46 uinput
crw-rw-rw- 1 root root 1, 9 Sep 28 04:46 urandom
crw-rw---- 1 root root 10, 257 Sep 28 04:46 userfaultfd
crw-rw---- 1 root tty 7, 0 Sep 28 04:46 vcs
crw-rw---- 1 root tty 7, 1 Sep 28 04:46 vcs1
crw-rw---- 1 root tty 7, 2 Sep 28 04:46 vcs2
crw-rw---- 1 root tty 7, 3 Sep 28 04:46 vcs3
crw-rw---- 1 root tty 7, 4 Sep 28 04:46 vcs4
crw-rw---- 1 root tty 7, 5 Sep 28 04:46 vcs5
crw-rw---- 1 root tty 7, 6 Sep 28 04:46 vcs6
crw-rw---- 1 root tty 7, 128 Sep 28 04:46 vcsa
crw-rw---- 1 root tty 7, 129 Sep 28 04:46 vcsa1
crw-rw---- 1 root tty 7, 130 Sep 28 04:46 vcsa2
crw-rw---- 1 root tty 7, 131 Sep 28 04:46 vcsa3
crw-rw---- 1 root tty 7, 132 Sep 28 04:46 vcsa4
crw-rw---- 1 root tty 7, 133 Sep 28 04:46 vcsa5
crw-rw---- 1 root tty 7, 134 Sep 28 04:46 vcsa6
crw-rw---- 1 root tty 7, 64 Sep 28 04:46 vcsu
crw-rw---- 1 root tty 7, 65 Sep 28 04:46 vcsu1
crw-rw---- 1 root tty 7, 66 Sep 28 04:46 vcsu2
crw-rw---- 1 root tty 7, 67 Sep 28 04:46 vcsu3
crw-rw---- 1 root tty 7, 68 Sep 28 04:46 vcsu4
crw-rw---- 1 root tty 7, 69 Sep 28 04:46 vcsu5
crw-rw---- 1 root tty 7, 70 Sep 28 04:46 vcsu6
drwxr-xr-x 2 root root 60 Sep 28 04:46 vfio
crw-rw---- 1 root root 10, 256 Sep 28 04:46 vga_arbiter
crw-rw---- 1 root kvm 10, 238 Sep 28 04:46 vhost-net
crw-rw---- 1 root kvm 10, 241 Sep 28 04:46 vhost-vsock
drwxr-xr-x 2 root root 100 Sep 28 04:46 virtio-ports
crw-rw---- 1 root root 243, 0 Sep 28 04:46 vport0p0
crw-rw---- 1 root root 243, 1 Sep 28 04:46 vport0p1
crw-rw---- 1 root root 243, 2 Sep 28 04:46 vport0p2
crw-rw-rw- 1 root root 10, 259 Sep 28 04:46 vsock
crw-rw-rw- 1 root root 1, 5 Sep 28 04:46 zero
alurusaranya@LAPTOP-0PA61QEN:~/practical9$

```

Practical 9 — Theory

Linux File System and VFS

The **Linux file system** is used to organize and store files and directories. Linux treats files and directories using a common file-system interface.

An **inode** is a data structure that stores information about a file, such as its permissions, owner, size, timestamps, and links. The filename is associated with the inode through the directory structure.

The `stat` command is used to display detailed information about a file, including its inode number, size, permissions, and timestamps.

The **Virtual File System (VFS)** provides a common interface between user programs and different file-system implementations. It allows Linux to work with different types of file systems through common operations.

Commands such as `df`, `mount`, `findmnt`, and `/proc/filesystems` can be used to examine mounted file systems, available storage, and file-system types.

Important commands

`ls -li`

`stat test.txt`

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`df -hT`

`findmnt .`

`cat /proc/filesystems`

`cat /proc/mounts`

Result

The practical demonstrates the basic structure of the Linux file system, inode information, mounted file systems, and the role of VFS in providing a common interface for file-system operations.